How BARCS Finds Important Genes

Chapter 1: The ideas

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Companion slides to the student guide · BARCS 0.2.1 · Grades 7 to 9

The plan for Chapter 1

1. The experiment: a CRISPR screen
2. The data
3. Step 1: Turn counts into proportions
4. Step 2: Compare the groups with a fold change
5. Step 3: Odds and log-odds
6. Step 4: Two kinds of noise
7. Step 5: The design matrix

Remember

Each section of these slides has the same title as a section of the textbook. The examples use the same numbers.

The experiment: a CRISPR screen

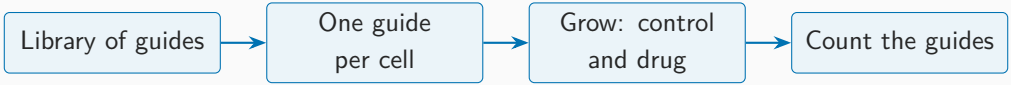
What scientists want to know

- Each cell has about 20,000 genes.
- A gene is a set of instructions.
- Question: which genes does a cell need, for example to survive a cancer drug?

A CRISPR screen tests thousands of genes at the same time.

A CRISPR screen in four steps

1. Make a library of short pieces of RNA, called guides. Each guide switches off one gene.
2. Put one guide into each cell. Now each cell has one gene switched off.
3. Grow the cells: some in normal conditions (control), some with a drug (drug).
4. A sequencing machine counts each guide. This number is a read count.



Why a count tells us about a gene

- Suppose cells need a gene to survive the drug.
- Cells that lose that gene die.
- So their guide becomes rare in the drug group.

Remember

BARCS does not look at cells directly. It looks at read counts. A guide that becomes rare tells us that its gene is important.

Replicates and libraries

- Scientists repeat each group more than one time. Each repeat is a replicate.
- Each sample that the machine reads is a library.
- Our example has four libraries:
  - two control replicates
  - two drug replicates

The data

Read counts for one guide

Library	Group	Guide count K	Total count N
1	control	180	1,000,000
2	control	240	1,200,000
3	drug	120	1,000,000
4	drug	80	800,000

We use these numbers on every slide.

What the symbols mean

K: the read count of one guide in one library (library 1: $K = 180$). N: the total read count of one library (library 1: $N = 1{,}000{,}000$). i: the library number ($K_2 = 240$).

Record the totals first

Remember

Always record N *before* you remove any guides from the table.

If you remove guides first, the totals become wrong. Then all the later numbers become wrong too.

Step 1: Turn counts into proportions

Divide each count by its library total

- Library 2 has more reads only because the machine read it more.
- So we cannot compare raw counts.
- Divide each count by its total. The result is a proportion:

$$p = \frac{K}{N}$$

- Proportions are tiny. Multiply by one million: per million reads.

What the symbols mean

p: the proportion of all reads that belong to the guide. Always between 0 and 1.

Worked example: proportions

Worked example 1

Library 1: $180 \div 1{,}000{,}000 = 0.000180 = 180$ per million
Library 2: $240 \div 1{,}200{,}000 = 0.000200 = 200$ per million
Library 3: $120 \div 1{,}000{,}000 = 0.000120 = 120$ per million
Library 4: $80 \div 800{,}000 = 0.000100 = 100$ per million

- Control: 180 and 200 per million.
- Drug: 120 and 100 per million.
- The guide is less common with the drug.

Step 2: Compare the groups with a fold change

Combine the replicates of each group

Add the counts and add the totals of each group:

$$p_{\text{control}} = \frac{180 + 240}{1{,}000{,}000 + 1{,}200{,}000} = \frac{420}{2{,}200{,}000} = 0.000191 \quad \text{(191 per million)}$$

$$p_{\text{drug}} = \frac{120 + 80}{1{,}000{,}000 + 800{,}000} = \frac{200}{1{,}800{,}000} = 0.000111 \quad \text{(111 per million)}$$

The fold change

$$\text{fold change} = \frac{p_{\text{drug}}}{p_{\text{control}}} = \frac{0.000111}{0.000191} = 0.58$$

The guide keeps only 58% of its control level. It loses 42%.

Remember

- Fold change $= 1$: no change.
- Fold change < 1: the guide becomes rarer. Cells may need this gene.
- Fold change > 1: the guide becomes more common. Losing this gene may help the cells.

Logarithms make changes symmetric

- "Double" is a fold change of 2. "Half" is 0.5.
- Same size of change, but 2 and 0.5 do not look equal.
- The base-2 logarithm, $\log_2$, counts how many times you double.

Fold change	0.25	0.5	1	2	4
$\log_2$(fold change)	-2	-1	0	1	2

Now "double" is $+1$ and "half" is -1.

Worked example: log fold change

Worked example 2

Our fold change is 0.58, between 0.5 and 1. So $\log_2(0.58)$ is between -1 and 0.
A calculator gives $\log_2(0.58) = -0.78$: the guide drops by 0.78 "halvings".

- BARCS uses the natural logarithm, ln: powers of $e = 2.718\ldots$ instead of powers of 2.
- To go back: if $\ln(x) = b$, then $x = e^b$.
- For our guide: $\ln(0.58) = -0.54$.

Step 3: Odds and log-odds

Odds and log-odds

The odds compare the reads of the guide with all other reads:

$$\text{odds} = \frac{p}{1-p}$$

The log-odds is the natural logarithm of the odds:

$$\text{log-odds} = \ln\left(\frac{p}{1-p}\right)$$

Worked example: odds and log-odds

Worked example 3

Control: $p = 0.000191$, so odds $= 0.000191 \div 0.999809 = 0.000191$.

For small p, the odds and the proportion are almost equal, because $1 - p$ is almost 1.

Log-odds, control: $\ln(0.000191) = -8.56$

Log-odds, drug: $\ln(0.000111) = -9.10$

Difference: $-9.10 - (-8.56) = -0.54$

The difference is the same as ln(fold change) from Step 2.

Why use log-odds?

Remember

For guides, a difference in log-odds is almost exactly the natural log of the fold change. So you can read BARCS results as log fold changes.

- A proportion must stay between 0 and 1.
- A log-odds can be any number, positive or negative.
- So BARCS can add effects together safely (Step 5).

Step 4: Two kinds of noise

Replicates are never equal

- The two control libraries gave 180 and 200 per million.
- This difference is noise.
- BARCS must know the size of the noise to tell a real change from a chance change.

1. Counting noise
The machine reads a random sample. Like a handful from a bag of a million marbles: the number of red marbles changes each time.

2. Biological noise
Each replicate is a different group of living cells. The cells grow a little differently each time.

Variance: the plan

- Variance: the average squared distance from the expected value.
- Standard deviation: the square root of the variance, in the units of the counts.
- The sign $\times$ means "multiply". $N \times \mu$ is "N times μ". It is not one symbol.

We build the formula in four parts: (1) the expected count, (2) counting noise, (3) biological noise, (4) add the two parts.

What the symbols mean

N: library total (1,000,000). μ ("mu"): expected proportion (control: 0.000191). ρ ("rho"): size of the biological noise.

- The machine reads N reads.
- Each read belongs to our guide with the chance μ.

$$\text{expected count} = N \times \mu$$

Worked example 4a

$N \times \mu = 1{,}000{,}000 \times 0.000191 = 191$ reads.

The real counts are near 191, but not exactly 191. The variance tells how far they normally are.

Part 2: One read — five marbles

One red marble (value 1), four blue (value 0). Chance of red $\mu = 0.2$. Average $= 0.2$.

Marble	Value	Distance	Squared
red	1	$+0.8$	0.64
blue	0	-0.2	0.04
blue	0	-0.2	0.04
blue	0	-0.2	0.04
blue	0	-0.2	0.04
Sum		0	0.80
Average		0	**0.16**

- Plain distances cancel: the sum is always 0.
- Squares are never negative, so they do not cancel.
- Average squared distance $= 0.80/5 = 0.16$.
- This is the variance.

Why squares? Because squared spreads add up

Two fair coins (heads $= 1$, $\mu = 0.5$). Four equally likely results:

Coins	Total	Dist. from 1	Squared
T, T	0	-1	1
T, H	1	0	0
H, T	1	0	0
H, H	2	$+1$	1

	Squared	Size only
One coin	0.25	0.5
One + one	0.5	1.0
Total of two	0.5	0.5
Adds up?	yes	no

- We add a million reads (Part 2) and two kinds of noise (Part 4). We need a spread that adds up.
- R. A. Fisher named this the *variance* in 1918 for this reason. Squared distances go back to Legendre's "least squares" (1805).

The general formula for one read

Replace "one in five" by any chance μ:

Value	How often	Distance from μ	Squared distance
1	μ	$1-\mu$	$(1-\mu)^2$
0	$1-\mu$	$-\mu$	μ^2

Average squared distance = (squared distance × how often), added:

$$\text{variance of one read} = \mu \times (1-\mu)^2 + (1-\mu) \times \mu^2$$

Worked example 4b: five marbles again

$0.2 \times 0.8^2 + 0.8 \times 0.2^2 = 0.128 + 0.032 = 0.16$ — the same as the table.

Open the square

A square means "multiply by itself":

$$(1-\mu)^2 = (1-\mu) \times (1-\mu)$$

Multiply each part of the first bracket by each part of the second:

$$(1-\mu) \times (1-\mu) = 1 \times 1 \; - \; 1 \times \mu \; - \; \mu \times 1 \; + \; \mu \times \mu = 1 - 2 \times \mu + \mu^2$$

Worked example 4b: check with numbers

With $\mu = 0.2$: $0.8 \times 0.8 = 0.64$ and $1 - 2 \times 0.2 + 0.04 = 0.64$.

From the long formula to $\mu \times (1 - \mu)$

With letters	With $\mu = 0.2$	What we did
1. $\mu \times (1-\mu)^2 + (1-\mu) \times \mu^2$	$0.128 + 0.032$	start
2. $\mu \times (1 - 2 \times \mu + \mu^2) + (1-\mu) \times \mu^2$	$0.128 + 0.032$	open the square
3. $(\mu - 2 \times \mu^2 + \mu^3) + (\mu^2 - \mu^3)$	$(0.2 - 0.08 + 0.008) + (0.04 - 0.008)$	multiply in
4. $\mu - 2 \times \mu^2 + \mu^2 + \mu^3 - \mu^3$	$0.2 - 0.08 + 0.04 + 0.008 - 0.008$	sort the powers
5. $\mu - \mu^2$	$0.2 - 0.04 = 0.16$	combine
6. $\mu \times (1 - \mu)$	$0.2 \times 0.8 = 0.16$	take μ out

Every line gives 0.16. Shorter way: take out the common part $\mu \times (1 - \mu)$:

$$\mu \times (1-\mu) \times \big[(1-\mu) + \mu\big] = \mu \times (1-\mu) \times 1$$

Part 2: Counting noise for N reads

- The reads are independent. Variances of independent values add up.
- N reads: N times the variance of one read.

$$\text{counting noise} = N \times \mu \times (1 - \mu)$$

Worked example 4c: many reads

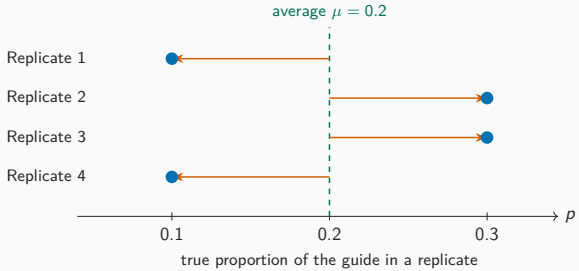
100 reads, $\mu = 0.2$: expected $100 \times 0.2 = 20$; variance $100 \times 0.16 = 16$; SD $\sqrt{16} = 4$.

Ten repeats gave 17, 19, 21, 25, 17, 25, 27, 22, 21, 14: near 20, about 4 away.

Our guide: $1{,}000{,}000 \times 0.000191 \times 0.999809 = 191$.

Part 3: What is the wobble?

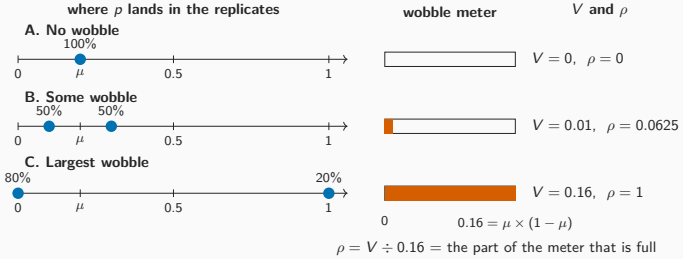
- Each replicate is different living cells. So the true proportion p of the guide is a little different in each replicate.
- The wobble is how far p moves from its average μ: V = average of $(p - \mu)^2$.



distance	squared
-0.1	0.01
$+0.1$	0.01
$+0.1$	0.01
-0.1	0.01
average:	$V = 0.01$

No wobble: every replicate at $p = 0.2$, all arrows 0, $V = 0$.

ρ: how full is the wobble meter?



- p must stay between 0 and 1. The widest wobble: p jumps to 0 or 1 (case C). That is the one-read case of Part 2: $V = \mu \times (1 - \mu)$.
- So $\mu \times (1 - \mu)$ is the full meter, and $\rho = V \div \big(\mu \times (1 - \mu)\big)$ is how full it is.
- Next two slides: turn this around to $V = \rho \times \mu \times (1 - \mu)$, then put it into the variance formula.

Turn the formula around

Multiply both sides by $\mu \times (1 - \mu)$:

$$\rho = \frac{V}{\mu \times (1 - \mu)} \quad \Longrightarrow \quad V = \rho \times \mu \times (1 - \mu)$$

Remember

ρ is not replaced by μ. The wobble = how full the meter is $\times$ the size of the full meter. ρ removes the size of the meter, so it means the same for a rare and a common guide.

Worked example 4d

Case B: $0.0625 \times 0.2 \times 0.8 = 0.0625 \times 0.16 = 0.01 = V$.

Our guide: full meter $= 0.000191 \times 0.999809 = 0.000191$; $\rho = 0.000000386$ (almost empty); $V = 0.000000386 \times 0.000191 = 0.0000000000737$; $\sqrt{V} = 8.6$ per million.

How ρ gets into the variance formula

Step	Formula	Case B, $N = 10$
1. Wobble	$V =$ average of $(p-\mu)^2$	$V = 0.01$
2. Meter	$\rho = V \div \big(\mu \times (1-\mu)\big)$	$0.01 \div 0.16 = 0.0625$
3. Turn around	$V = \rho \times \mu \times (1-\mu)$	$0.0625 \times 0.16 = 0.01$
4. Part 4	$N \times \mu \times (1-\mu) + N \times (N-1) \times V$	$1.6 + 90 \times 0.01 = 2.5$
5. Replace V	$N \times \mu \times (1-\mu) + N \times (N-1) \times \rho \times \mu \times (1-\mu)$	$1.6 + 90 \times 0.0625 \times 0.16 = 2.5$
6. Take out	$N \times \mu \times (1-\mu) \times \big(1 + (N-1) \times \rho\big)$	$1.6 \times 1.5625 = 2.5$

$N \times \mu \times (1-\mu) = 10 \times 0.2 \times 0.8 = 1.6$; $\quad N \times (N-1) = 10 \times 9 = 90$; $\quad 1 + 9 \times 0.0625 = 1.5625$.

Step 6 takes out the common part $N \times \mu \times (1-\mu)$. In step 5, the orange V is replaced by $\rho \times \mu \times (1-\mu)$ from step 3. Every line gives the same number.

Part 4: Split the distance into two pieces

In a replicate with true proportion p:

$$K - N \times \mu = \underbrace{(N \times p - N \times \mu)}_{\text{piece 1: the wobble}} + \underbrace{(K - N \times p)}_{\text{piece 2: counting noise}}$$

- Piece 2 goes up or down equally, whatever p is. The pieces are unrelated, so their variances add (Part 2, Step B).
- **Piece 1:** multiply a value by $N \Rightarrow$ distances $\times N \Rightarrow$ squared distances $\times N \times N$.
  Example: 1, 2, 3 have variance 0.667; 10, 20, 30 have variance 66.7 ($\times 100$).

$$\text{variance of piece 1} = N \times N \times V$$

Piece 2: counting noise is a little smaller

Counting noise for one replicate is $N \times p \times (1-p)$. Average it:

With letters	Case B ($p = 0.1$ or 0.3)
1. avg. of $p^2 = \mu^2 + V$	$(0.01 + 0.09)/2 = 0.05 = 0.04 + 0.01$
2. avg. of $p \times (1-p) = \mu -$ avg. of p^2	$(0.09 + 0.21)/2 = 0.15$
3. $= \mu - (\mu^2 + V)$	$0.2 - 0.05 = 0.15$
4. $= \mu \times (1-\mu) - V$	$0.16 - 0.01 = 0.15$

$$\text{variance of piece 2} = N \times \mu \times (1-\mu) - N \times V$$

Line 1: open $(p-\mu)^2$; the average of p is μ. A wobbling p spends time near the edges, where counting noise is smaller.

Where $N \times (N-1)$ comes from

$$\begin{aligned}\text{variance} &= N \times N \times V + N \times \mu \times (1-\mu) - N \times V \\ &= N \times \mu \times (1-\mu) + (N \times N - N) \times V \\ &= N \times \mu \times (1-\mu) + N \times (N-1) \times V\end{aligned}$$

Remember

The wobble adds $N \times N \times V$ but takes away $N \times V$ from counting noise. What remains is $N \times (N-1) \times V$.

Put in $V = \rho \times \mu \times (1-\mu)$ and take out $N \times \mu \times (1-\mu)$:

$$\text{variance} = \underbrace{N \times \mu \times (1-\mu)}_{\text{counting noise}} \times \underbrace{(1 + (N-1) \times \rho)}_{\text{extra factor}}$$

Check with ten reads

Worked example 4e

Case B ($\mu = 0.2$, $V = 0.01$, $\rho = 0.0625$) with $N = 10$:

Piece 1:	$10 \times 10 \times 0.01 = 1.0$
Piece 2:	$10 \times (0.16 - 0.01) = 1.5$
Total:	$1.0 + 1.5 = \mathbf{2.5}$
Boxed formula:	$1.6 \times (1 + 9 \times 0.0625) = 1.6 \times 1.5625 = \mathbf{2.5}$
Computer, 100,000 replicates:	2.498
With $N \times N$ instead:	$1.6 + 100 \times 0.01 = 2.6$ (wrong)

Worked example: total variance of our guide

Worked example 4f

Piece 1:	$1{,}000{,}000 \times 1{,}000{,}000 \times 0.0000000000737 = 73.7$
Piece 2:	$191.0 - 0.00007 = 191.0$
Total:	$73.7 + 191.0 = 264.7$
Boxed formula:	$191 \times (1 + 999{,}999 \times 0.000000386) = 191 \times 1.386 = 264.7$
Standard deviation:	$\sqrt{264.7} = 16.3$ reads

For a million reads the -1 changes almost nothing. Counting noise alone gives $\sqrt{191} = 13.8$. Biological noise adds 39% to the variance.

More reads do not remove biological noise

Remember

The extra factor $1 + (N - 1) \times \rho$ contains N.

When the machine reads more, N grows, and the factor grows too.

So more reading does not remove biological noise. Only more replicates help.

Step 5: The design matrix

What a design matrix is

A design matrix X is a table of numbers:

- one row for each library
- one column for each β value

Lib.	Group	Intercept	Drug
1	control	1	0
2	control	1	0
3	drug	1	1
4	drug	1	1

What the symbols mean

Intercept: a column of 1s. It belongs to β_0, the starting level.

Drug: 1 for a drug library, 0 for control. It belongs to β_1.

β_0: the control log-odds.

β_1: the change from control to drug. This is the effect.

From the design matrix to a prediction

Take the row of a library. Multiply each number by its β. Add:

$$\eta = (\text{column 1}) \times \beta_0 + (\text{column 2}) \times \beta_1$$

Worked example 5: predictions

BARCS finds $\beta_0 = -8.565$ and $\beta_1 = -0.542$ (Step 6 in Chapter 2).

Libraries 1, 2 (row 1, 0): $1 \times (-8.565) + 0 \times (-0.542) = -8.565$

Libraries 3, 4 (row 1, 1): $1 \times (-8.565) + 1 \times (-0.542) = -9.107$

Control gets β_0. Drug gets $\beta_0 + \beta_1$. So β_1 is the difference.

From log-odds back to counts

$$\mu = \frac{e^{\eta}}{1 + e^{\eta}}, \qquad \text{expected count} = N \times \mu$$

Worked example 6: expected counts

Control: $e^{-8.565} = 0.000190676$, so $\mu = 0.000190640$ (190.6 per million). Drug: $\mu = 0.000110827$ (110.8 per million).

Library	N	μ (per million)	Expected $N \times \mu$	Real K
1	1,000,000	190.6	190.6	180
2	1,200,000	190.6	228.8	240
3	1,000,000	110.8	110.8	120
4	800,000	110.8	88.7	80

The differences (-10.6, $+11.2$, $+9.2$, -8.7) are the noise.

The reference group

- Control has 0 in the drug column. So control is the reference group.
- β_1 always compares a group with the reference group.
- If drug becomes the reference group, β_1 changes sign: $+0.542$. The meaning stays the same.

Paired replicates: add a column

If replicate 1 of control and drug come from the same batch, add a batch column:

Lib.	Rep.	Int.	Rep. 2	Drug
1	1	1	0	0
2	2	1	1	0
3	1	1	0	1
4	2	1	1	1

Each new column costs one degree of freedom.

Worked example 7: the paired model

Drug effect: $\beta = -0.544$, SE 0.143.
Degrees of freedom: $4 - 3 = 1$ (was 2).
p-value: 0.164 (was 0.033).
The batch difference is only 0.005. The column costs information and gives nothing back.

Numbers instead of groups: a time course

Lib.	Day	Int.	Day ÷ 14
1, 2	0	1	0
3, 4	7	1	0.5
5, 6	14	1	1

β_1 is the change in log-odds over 14 days.

Worked example 8: a time course

Suppose $\beta_1 = -0.70$.

Day 14: $1 \times (-0.70) = -0.70$; $e^{-0.70} = 0.50$. Half.

Day 7: $0.5 \times (-0.70) = -0.35$; $e^{-0.35} = 0.70$. 70%.

Degrees of freedom: $6 - 2 = 4$.

Two factors and an interaction

Group	Intercept	Knockout	Drug	Knockout × Drug
normal, no drug	1	0	0	0
knockout, no drug	1	1	0	0
normal, drug	1	0	1	0
knockout, drug	1	1	1	1

- The last column is knockout $\times$ drug. It is 1 only when both are 1.
- Its β is the interaction: the extra change that happens only when both changes happen together.

Worked example: reading an interaction

Worked example 9

The four β values are -8.5, -0.2, $+0.1$ and -0.7.

normal, no drug: -8.5

knockout, no drug: $-8.5 + (-0.2) = -8.7$

normal, drug: $-8.5 + 0.1 = -8.4$

knockout, drug: $-8.5 + (-0.2) + 0.1 + (-0.7) = -9.3$

If the effects only added up: $-8.5 - 0.2 + 0.1 = -8.6$. The real value is -9.3. The extra -0.7 is the interaction.

Comparing two treatments: a contrast

- Three groups: control, drug A, drug B.
- β_A compares drug A with control. β_B compares drug B with control.
- To compare drug A with drug B, subtract: $\beta_A - \beta_B$. This is a contrast.
- In BARCS: `bb_contrast()`.

Example: $\beta_A = -0.9$, $\beta_B = -0.4$, so $\beta_A - \beta_B = -0.5$. Drug A lowers the guide more.

Rules for a good design matrix

1. One row for each library, in the order of the count table.
2. One column for each β that you want.
3. More rows than columns. The difference is the degrees of freedom. Two or more is better than one.
4. No two columns with the same information.

Remember

Suppose all drug libraries come from batch 2 and all control libraries from batch 1. Then the batch column equals the drug column. BARCS cannot separate the drug effect from the batch effect.

The design matrix in R

A formula describes X. `model.matrix()` builds it:

```
> data <- data.frame(group = c("control", "control", "drug", "drug"))
> model.matrix(~ group, data)
  (Intercept) groupdrug
1           1         0
2           1         0
3           1         1
4           1         1
```

Then give BARCS the counts, totals, formula and column name:

```
fit <- bb_screen(counts, data, formula = ~ group,
                 term = "groupdrug", totals = totals)
```

Paired: `~ replicate + group`. Interaction: `~ knockout * drug`.

Chapter 1 summary

What Chapter 1 showed

1. BARCS turns read counts into proportions (Step 1).
2. It compares groups with a fold change and its logarithm (Step 2).
3. It works on the log-odds scale (Step 3).
4. It separates counting noise from biological noise (Step 4).
5. It describes the experiment with a design matrix (Step 5).

Chapter 2 shows how BARCS calculates the effect and decides if it is real.

The symbols of Chapter 1

Symbol	Meaning	Value in our example
K	read count of the guide in one library	180, 240, 120, 80
N	total read count of one library	1.0, 1.2, 1.0, 0.8 million
p	proportion, K/N	191 and 111 per million
fold change	$p_{\text{drug}} \div p_{\text{control}}$	0.58
$\log_2$, ln	doublings; powers of e	-0.78; -0.54
log-odds	$\ln\big(p/(1-p)\big)$	-8.56 (control)
μ	expected proportion	0.000191 (control)
ρ	biological noise	0.000000386
X	design matrix: a row per library, a column per β	4 rows, 2 columns
β_0, β_1	control log-odds; effect of the drug	-8.565; -0.542
η	predicted log-odds (row of X times β)	-8.565, -9.107